Step 1: Choose Dataset

Use something simple

Dataset : Student performance (Pass/Fail)

Study Hours	Attendance	Assignment	Result
High	High	Yes	Pass
High	Low	Yes	Pass
Medium	High	Yes	Pass
Low	Medium	No	Fail
Low	Low	No	Fail
Medium	Medium	Yes	Pass
High	Medium	Yes	Pass
Low	High	No	Fail

Step 2: Entropy of Dataset

Entropy (S) $= -P_1 \log_2 P_1 - P_2 \log_2 P_2$

* Pass = 5
* Fail = 3

$$\text{Entropy}(S) = -\frac{5}{8}\log_2\frac{5}{8} - \frac{3}{8}\log_2\frac{3}{8}$$

Final ≈ 0.954

Step 3: Calculate for Each Feature

* Example : Study Hours

Split

* High → (3 Pass, 0 Fail) → Entropy = 0
* Medium → (2 Pass, 0 Fail) → Entropy = 0
* Low → (0 Pass 3 Fail) → Entropy = 0

→ Weighted Entropy = 0

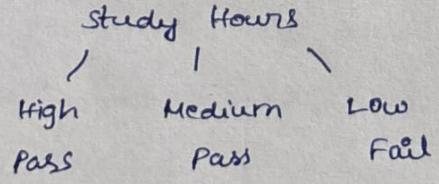
IG

$$IG = 0.954 - 0 = 0.954$$

Step 4: Root Node

* Feature with highest Information Gain = Study Hours

Step 5: First Level tree



Part 2 : CART

Step 1 : Gini Index Formula

$Gini(S) = 1 - \sum P_i^2$

Dataset :

$$Gini = 1 - (5/8)^2 - (3/8)^2 = 0.468$$

Step 2 : Choose Feature

Binary split

* Split : {High, Medium} vs {Low}

Step 3 : Calculate Gini

Left Node (High + Medium) :

* 5 Pass, 0 Fail → Gini = 0

Right Node (Low) :

* 0 Pass, 3 Fail → Gini = 0

Step 4 : weighted Gini

$$= (5/8) * 0 + (3/8) * 0 = 0$$

Step 5 : Gini Reduction

$$0.468 - 0 = 0.468$$

Step 6 : Theory

CART always produce binary tree because it split data into two groups at each step. This makes computation simpler and more efficient. Binary split also help in handling both numerical and categorical, data efficiency.